

Advancing Insider Threat Detection with Unsupervised Anomaly Detection and User Behavior Analytics

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Abstract — Insider threats are among the hardest to manage when it comes to protection of an organization from cyber threats since the insiders are authorized users with access to organizational resources. Identifying such threats require sophisticated processes in anomaly detection and behaviour analytics to pinpoint such acts. In this article, we present a methodology which uses machine learning paradigm to model activity profiles of the users as well as to identify ‘abnormal behavior’ that indicates susceptibility to insider threat. Our approach incorporates both clustering and autoencoder techniques to identify user activity baselines from the unsupervised data. To measure the potential of these methods for detecting anomalous patterns indicating malicious actions, we assess how they work using real-world datasets. The outcomes underline the ability of automated behavior analysis in improving the UEBA cybersecurity frameworks and suggest future improvements in making UEBA solutions more affordable and effective.

Keywords: Insider threats, anomaly detection, user behavior analytics (UBA), autoencoders

I. INTRODUCTION

The advent of digital devices has brought insane changes to our lifestyle. Businesses, organisations, and governments have adapted their operations to modern digital systems. Several ERP, CRM and word processing software have been developed which have digitised business operations. Such an increased reliance on these digital systems have brought security challenges like data and infrastructure security, cyberattack vulnerabilities, privacy breaches, unauthorised access. Cyber security is an active research area to study such threats and vulnerabilities with their detections and mitigation. **Various tools are developed both open source (like WAZUH) and propriety (like)** to detect and mitigate cyber threats and to keep the organisation and their infrastructure safe. These threats are posed by entities outside of the organisation such as an unauthorised person accessing an organisation's data servers, which can be easily identified if

proper tools and planning is employed. However, if the attacker is within the organisation, i.e. he has legitimate access to the organisation's infrastructure, it makes the system insanely vulnerable as it is nearly impossible to identify the threat.

A 2018 insider threat report finds that 53% of the surveyed organisations reported insider attacks and 27% reports that the attacks are more frequent[1]. This is because insider threats are particularly challenging to detect, as attackers often have legitimate access to an organisation's resources. According to cisa.gov, “insider threat is the potential for an insider to use authorised accesses or organisation's knowledge to harm that organisation”[2] For example, an employee with access to sensitive financial data might leak it to competitors for personal gain, or a disgruntled staff member could sabotage internal systems, causing operational disruptions. Insiders could also unknowingly compromise security by falling victim to phishing schemes, providing cybercriminals with access to critical resources through legitimate credentials. These threats are particularly dangerous because insiders are often trusted, making detection more difficult.

Given the complexity and significance of insider threats, the need for more dedicated and innovative approaches in anomaly detection and user behaviour analytics (UBA) has emerged. Behaviour analytics focuses on comprehending the past and present behaviour of entities and based on this predicts the future behaviour using ML[3] The domain of behaviour analytics has found its applications in various domains like consumer behaviour analysis, bus line design optimizations [4], product recommendations [5] only few to mention. In the case of insider threat detection, the users' behaviour is monitored to create a baseline for their normal activities. Statistical models are trained on this baseline and any deviation from the established behaviour triggers an alert.

Currently there is no dedicated open source UEBA solution, while only a few enterprise-grade vendors offer comprehensive UEBA tools. A significant research has been conducted in the field of behavioural analytics, but there has been a lack of corresponding development in practical, accessible solutions. As a result, the market for UEBA solutions is not highly competitive, with few tools available to address the growing need for sophisticated behaviour-based security monitoring. This gap presents an opportunity for innovation and the development of new, more accessible UEBA platforms that can meet the demands of modern security landscapes.

In this paper, we present a novel method to handle insider threat detection by using autoencoders, particularly using anomaly detection techniques under unsupervised learning. Thus, using autoencoders we intend to learn the regular users' activity patterns in an organization and detect the security threats as differences from these patterns. The autoencoder network is learnt to estimate patterns of user behaviors from past activity logs, and the residual error is used in identifying misbehavior. This method of decision making allows for the identification of odd, precursory behaviors that may be quite hard to define in rule-based systems. Due to the fact that we're searching for malicious actions from insiders that already have authorized access to the organization's hardware and software, our approach fits this particular type of intelligence gathering. And where autoencoders are applicable, they allow detecting such anomalous patterns without the necessity to label data, which will make it possible to propose efficient and scalable real-time monitoring of user activities.

II. RELATED WORK

User and entity behavior analytics (UEBA) has been emerging as a further area of interest in cybersecurity literature especially for insider threat. In recent years a number of approaches based on machine learning and statistics have been tried for detection of intrusions on user activity logs with reasonable to high successes in real-world setting.

Martin Ussath et. al. [6] reviewed the neural networks for malicious user behavior detection using feed-forward neural network and recurrent neural network. Such work raises important issues that concern the degree to which accuracy can depend on the complexity of the model, the quality of the data set, and the number of iterations. In more detail, the increase of models complexity and the availability of richer data sets increased the models accuracy up to 98% for FNN and up to 97% for RNN. Nevertheless, this study was confined to focusing on synthetic environments

using synthetically generated datasets in order keep the environmental complexity to the minimal level. Further, aspects like computational constraints, and other real-time application of the model were kept as under-explored, which leads to the ambiguity of implementing these models.

Ahmed et al. [7] extended the current work by converting user activity logs into the problem of time series analysis with the help of matrix profiling. This way they are also can scale their system based on the GPU accelerated computation to manage large data set. Their non-parametric approach helped them avoid overloading on the false positives because they did a comparison of overlapping anomalous events arising from features and sub-features, which assisted in improving their detection function. However, its vulnerability to empirically tuned parameters, the size of window, for instance, implies some difficulties in the transfer of the discussed approach to other datasets with divergent user behaviors.

Sharma et al. [8] describe another approach based on more integration of Splunk User Behavior Analytics and the Securonix Security Analytics Platform. This combined framework employed automated machine learning, behavioral modeling, and risk scoring to prioritize threat agents and forms a single integrated system for threat identification. Although the integration of such platforms increases threat detection, the practicality of implementing such systems for operational use is complicated by increasing complexity of organizations and scalability in different environments.

Other types of RNN have also been used for the identification of behavior that is malicious in web access logs as pointed out by Ando et al. [9] They were well equipped to handle some of the challenges involved in work with time-series data by implementing the Long Short-Term Memory (LSTM) units, which worked to reduce the gradient problems as well. The results confirmed that LSTM based RNN had superior performance over conventional models such as Support Vector Machines (SVMs) in identifying fraudulent transactions. However, the study was hampered by the usual problem of class imbalance in security data sets, thus constraining the extent of generalization on new data. Solving this problem remains an interesting avenue for future work to guarantee that models' performance is optimized when it comes to imbalanced datasets, which are common in most security-related datasets.

Prarthana et al. [10] proposed a multidimensional data analysis framework for anomaly detection to be applied to OLAP cubes. Their approach facilitated categorization of event logs into the star schema besides permitting the use of statistical and machine learning techniques for dynamic anomaly detection approach. While this method was easy to

use and offered interpretations by visual interfaces, it could not be helpful in real-time processing of large volumes of data. However, the use of visual tools to analyze the data for presenting it may need to be better enhanced to make the tool friendly to non-professionals.

Similarly, Chattopadhyay et al. [11] aimed at using the deep autoencoder neural networks for performing time-series analysis in order to identify insider threats. They were able to monitor small changes in the overall behavior as well, not just extreme incidents due to the fact that they built time-series features including the first logon of the day and number of devices used per day. This approach proved to show great capability in detecting insider threats, better than the conventional models such as the random forest. They adopted balanced datasets and appropriate time windows to enhance accuracy; however, the accuracy of the method relies on the ability to find appropriate window sizes for every type of threat.

Nikiforova et al. [12] analyzed an approach of insider threat detection based on the clustering method with the use of K-means for group user behavior. Applying the partitioning clustering algorithms they resolved to identifying several outliers based on user behavior by comparing it with others in the group. The authors showed that their design was effective and accurate in detecting the anomalous users particularly when applied in large datasets. However it was restricted in its way to confirm these triggers as part of insider threats as the system could not directly affirm malicious acts. In addition the proposed did not consider change in the user behavior in the process of using the apps and it also did not compare its functionality against other malicious actions.

Overall, these works together establish the necessity for fine-tuning that challenges the anomaly detection models to networks, temporal models, or clustering schemes in regards to the given problem domain. With popular methods displaying potential in solving it, overall obstacles like scalability, class imbalance, as well as real-time function, are still primary setbacks. Future work will be required to address these shortcoming for the development of more reliable and extensible insider threat detection mechanism.

Authors	Methodology	Key Strengths	Limitations	Future Directions
Martin Ussath et al. [6]	Feed-Forward Neural Network (FNN) and Recurrent Neural Network (RNN) for malicious user behavior detection.	High accuracy (up to 98% for FNN, 97% for RNN) with rich datasets.	Confined to synthetic datasets; real-world applicability and computational constraints not explored.	Expanding the work to real-world datasets and considering real-time application and computational resources.
Ahmed et al. [7]	Time-series analysis with matrix profiling (STOMP) and GPU-accelerated computation.	Scalable for large datasets; non-parametric approach reduces false positives.	Vulnerable to empirically tuned parameters (e.g., window size), which limits generalizability.	Developing more generalized parameter selection techniques for diverse datasets and environments
Sharma et al. [8]	Integration of Splunk UBA and Securonix Security Analytics Platform for threat detection and risk scoring.	Provides a unified system for threat detection and prioritization.	High complexity in system integration and scalability concerns in diverse operational environments.	Exploring more streamlined integration methods and ensuring scalability in dynamic environments.
Ando et al. [9]	Recurrent Neural Network (RNN) with LSTM units for detecting malicious behavior in web access logs.	LSTM-based RNN outperforms SVMs, handles time-series data effectively.	Class imbalance in security datasets limits generalizability.	Addressing class imbalance in security datasets to improve generalization to unseen data.

Prarthana et al. [10]	Multidimensional data analysis using OLAP cubes for anomaly detection.	Facilitates dynamic anomaly detection and offers visual interfaces for easier interpretation.	Struggles with real-time processing of large datasets; limited scalability.	Enhancing real-time capabilities and improving user interface for non-professionals.
Chattopadhyay et al. [11]	Time-series analysis using deep autoencoder neural networks for insider threat detection.	Effective in detecting insider threats; better than conventional models like random forests.	Relies on finding optimal window sizes, which can vary for different threats.	Improving window size selection techniques for different types of insider threats.
Nikiforova et al. [12]	K-means clustering for grouping users based on behavior and identifying anomalous behavior.	Effective in detecting outliers and works well with large datasets.	Cannot directly confirm whether detected anomalies are insider threats; does not handle changing behaviors.	Incorporating dynamic behavior changes and improving accuracy in distinguishing between anomalous and malicious actions.

Table 1 Qualitative analysis of the Related works

III. METHODOLOGY

In this section, we describe the steps followed to detect anomalous user behavior using unsupervised learning on the CERT r4.2 dataset. The process includes data selection, feature extraction, model preparation, and anomaly detection, with each step detailed below.

A. Dataset

We used the CERT Insider Threat Dataset (r4.2), a synthetic dataset simulating user behavior in a corporate environment, with the potential for insider threat activity. This dataset was developed by the CERT Division in collaboration with ExactData, LLC, and is widely used for cybersecurity

research. There are six releases of the CERT dataset, and we worked with version 2 of release 4 (r4.2), which contains multiple types of user activity logs captured over time, including logon activity, device usage, file access, and web browsing.

The dataset includes CSV files that capture various activities: logon/logout events, device connections, email communications, psychometric tests, web browsing, and file operations. Additionally, the LDAP directory files provide organizational information, such as user roles and departments. For our study, we extracted user role information from the LDAP directory to match each user's activities with their respective organizational position and job responsibilities.

While this dataset is labeled—indicating whether an activity is benign or malicious—we chose to ignore the labels to focus on unsupervised anomaly detection. Our analysis included only selected files: logon.csv, psychometric.csv, device.csv, and file.csv. User data from the LDAP directory was merged to create a list of unique users with associated attributes (user_id, role, department), which was saved as a separate CSV file for further use.

B. Feature Extraction

The CERT r4.2 dataset is a time-based dataset, meaning that each activity is associated with a timestamp. The raw features available in the dataset are numerous, so we focused on a set of relevant features for anomaly detection. These features were chosen based on their potential to reflect abnormal behavior in the context of insider threats.

We first filtered users belonging to the ResearchAndEngineering functional unit and the Research department. This was done to maintain homogeneity in user roles and to limit the analysis to a group with similar responsibilities, which ensures that any detected anomalies are not simply due to role-based differences. Additionally, focusing on this unit is crucial because intellectual property theft is more likely to occur in research environments.

To generate meaningful inputs for our model, we aggregated key temporal and behavioral features for each user:

- 1) *Logon/Logoff Features*: For each user, we calculated the minimum, maximum, mean, and mode timestamps of logon and logoff events. These capture patterns in work hours and can indicate suspicious behavior if anomalies arise (e.g., unusually late logons).
- 2) *Device Connection/Disconnection Features*: For each user, we calculated the same set of aggregated statistics (minimum, maximum, mean, mode) for

device connection and disconnection times. Additionally, we extracted the number of file transfers associated with each connection, which is important for identifying abnormal data transfer activities.

- 3) *PC Usage Count*: We counted the number of distinct PCs accessed by each user, since excessive or unfamiliar device access might indicate a compromised account.
- 4) *Psychometric Features (OCEAN)*: For each user, we extracted psychometric attributes based on the OCEAN model (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism), which describes key personality traits. These features provide insight into the personality characteristics of users, potentially revealing patterns associated with higher insider threat risks.

Finally, we combined these features into a unified dataset for each user, merging the logon/logoff data with PC usage counts and *Psychometric* to investigate the combined effect of these variables.

C. Model Preparation

The objective of our study was to detect insider threats using unsupervised anomaly detection. We employed autoencoders, a type of neural network designed for unsupervised learning, to identify anomalous patterns in user behavior. Each feature set (logon/logoff, device connections, PC counts, and psychometric data) was trained separately with a unique instance of the autoencoder. This allowed us to assess anomalies in each behavioral dimension independently.

Before training, we used Min-Max Scaling to normalize the feature values between 0 and 1, ensuring that all variables contributed equally to the model. The Min-Max Scaling transformation for each feature x is given by:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the feature x in the training data.

The architecture of our autoencoder consisted of an encoder that progressively reduced the dimensionality of the input data and a decoder that reconstructed the data from this compressed representation. The idea was that the autoencoder would learn to reconstruct normal behavior well but fail to reconstruct anomalous behavior, resulting in a higher

reconstruction error for anomalies. The architecture can be summarized as:

1) *Encoder*:

$$h = \text{ReLU}(W_2 \cdot (\text{ReLU}(W_1 \cdot x + b_1)) + b_2)$$

2) *Decoder*:

$$\hat{x} = \text{Sigmoid}(W_4 \cdot (\text{ReLU}(W_3 \cdot h + b_3)) + b_4)$$

The network was trained using the Mean Squared Error (MSE) loss function, defined as:

$$\text{MSE}(x, \hat{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

where x represents the original input, and $\hat{x}$ is the reconstructed input from the autoencoder.

Training was done on 80% of the data, with 20% reserved for testing. The autoencoder was optimized using the Adam optimizer, with the training run for 100 epochs and a batch size of 32.

D. Anomaly Detection

After training, we used the autoencoder's reconstruction error to detect anomalies. For each input, the reconstruction error was calculated as the mean squared error (MSE) between the original input and the reconstructed data. The MSE was then transformed into a z-score to standardize the distribution:

$$z = \frac{\text{MSE} - \mu}{\sigma}$$

where μ is the mean of the MSE values, and σ is the standard deviation.

Anomalies were identified based on a threshold: any data point with a z-score greater than 2 standard deviations from the mean was considered anomalous. This method allowed us to flag users whose behavior significantly deviated from the norm, signaling potential insider threats.

IV. RESULTS AND DISCUSSION

After preprocessing and filtering, the CERT r4.2 dataset yielded 230 records for the combined logon/logoff and PC count data, 230 records for logon/off data alone, 66 records for device connect/disconnect paired with file transfer counts, and 230 records for PC access counts.

For each user, features were aggregated based on key statistics—minimum, maximum, mean, and mode values—reflecting their temporal behavior across different activities. From each feature set, 80% of the data was used for training, ensuring that the model learned from the majority of the user activity data while reserving the rest for testing.

A. Model Performance and Reconstruction Error

The autoencoder model was trained on these aggregated features to detect anomalous behavior by learning normal activity patterns and reconstructing the input data. The process begins by feeding in the training data, where the model compresses the input through a series of hidden layers (encoder), reducing it to a latent representation, and then reconstructs the data through a corresponding set of layers (decoder). The mean squared error (MSE) between the original input and the reconstructed output serves as the measure of how well the model has captured normal behavior. Lower reconstruction errors signify activities consistent with the model's learned patterns, whereas higher errors indicate potential anomalies.

Since this was an unsupervised learning task, the data lacked labels to denote normal or abnormal behavior. Therefore, we established thresholds for anomaly detection based on expert knowledge and an empirical analysis of the reconstruction errors. Specifically, the thresholds were set by visually inspecting the error distributions and selecting cutoffs that were one to 1.5 standard deviations away from the mean reconstruction error for each feature set. For instance, the threshold was set at 1 standard deviation for device connect/disconnect with file transfer count, 1.5 for logon/off and the combined parameters, and 1 for the PC access count.

B. Epoch Losses and Anomalies Detected

During training, the model's performance was monitored by tracking the training and validation loss for each feature set over 100 epochs. The convergence patterns, as illustrated in Fig. 1, reveal that the model successfully minimized reconstruction error across all feature sets over time, with the most significant decrease occurring in the first 20–30 epochs, followed by a plateau. The PC feature set showed the fastest convergence, with both the training and validation losses closely aligned, indicating good generalization to unseen data. Similarly, the logon/off feature set converged quickly, though with a slightly higher final loss compared to the PC set.

However, the device feature set and the combined feature set (all features aggregated) showed a more gradual reduction in loss, with the device data exhibiting the highest loss at the end of training. This suggests that the model had

more difficulty capturing device-related behaviors, likely due to the complexity or noisiness of the device usage patterns. Furthermore, the gap between training and validation losses for the device feature set was more pronounced, hinting at potential overfitting or the model's difficulty in generalizing device behavior to new, unseen data.

Once the training phase was complete, we applied the model to the test set, and anomalies were detected using Z-scores of the reconstruction errors for each feature set. As illustrated in Fig. 2, the PC feature set exhibited the highest density of anomalies, with multiple Z-scores exceeding the threshold, indicating significant outliers in user behavior related to PC activity. This suggests that there is greater variability in how users interact with PCs, possibly due to more complex or diverse PC usage patterns.

On the other hand, the device feature set showed the fewest anomalies, with most data points clustering within the normal range of Z-scores, suggesting that either device usage behavior is more predictable or that the model struggled to capture meaningful anomalies from device activity. The logon/off feature set revealed a few scattered anomalies, indicating some irregularities in user logon/off activities, though these were less frequent compared to PC-related anomalies.

C. Summary of Results

In summary, the PC and logon/off feature sets demonstrated the most robust performance in terms of model training and generalization, with faster loss convergence and lower overall reconstruction errors. The device feature set, in contrast, posed more challenges, with higher losses and fewer detected anomalies, suggesting either that device data is inherently more difficult to model or that it might require more sophisticated techniques or feature engineering. The Z-score analysis revealed that PC-related activities exhibited the greatest variance and were the most susceptible to anomalies, while device usage appeared to be more stable but harder to model effectively.

The thresholds set for anomaly detection—ranging from 1 to 2 standard deviations—allowed for a tailored identification of outliers in each feature set, with the PC feature set yielding the most anomalies, followed by logon/off activities and device usage. These findings highlight the distinct behavioral patterns in different types of user activities and underscore the need for customized anomaly detection approaches depending on the feature set.

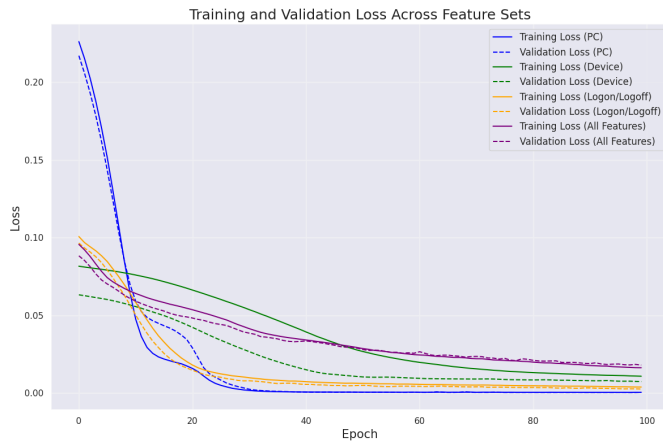


Fig. 1 Training and Validation Loss Across Feature Sets.

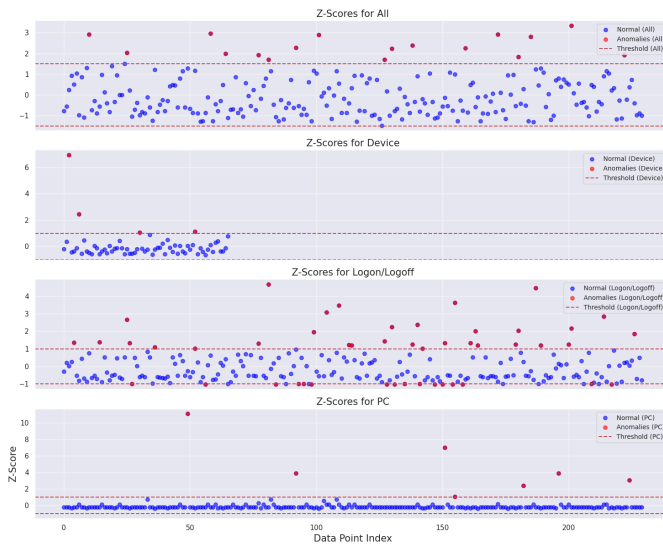


Fig. 2 Z-Scores of Reconstruction Errors for Anomaly Detection Across Feature Sets.

V. CONCLUSION

Altogether, this paper outlines a novel approach to the insider threat problem: the use of autoencoders for the purpose of anomaly detection on user activity profiles. The proposed technique reconstructs the normal usage patterns and marks the deviation from such as anti-pattern, and therefore the method presented here yields an efficient approach towards handling such anomaly without the need for training data. The results further show that autoencoders are capable of identifying multiple behavioral patterns within the features under consideration making it ideal for constant surveillance of activities of users within an organization. The use of this approach meets the need for higher behavior-based security solutions due to the constantly evolving insider threats. Since this study still needs further enhancement and testing on other benchmarks, the present work advances the creation of more

efficient and affordable UEBA systems for emerging cybersecurity threats.

VI. FUTURE WORK

In future research studies, several recommendations can be undertaken to improve on the proposed anomaly detection framework. First, trying to incorporate a larger set of user's activities along with other type of features such as network traffic, file access pattern, system log data and so on would involve a better understanding of both normal and anomalous user activities, hence more robust model. Further, including other types of Deep Learning, such as, Variational Autoencoder (VAE) or Generative Adversarial Network (GAN), could enhance the performance of the model in regard to perceive high dimensional data and small anomalies.

Another area that has a great potential for enhancement is the calibration of the thresholding mechanism for purposes of anomaly detection. Currently threat detection relies on expert-derived thresholds in the form of Z-scores; dynamic thresholding methods or unsupervised clustering can improve flexibility and scalability of threat identification. Incorporating the temporal aspects of users' behaviour patterns, including time dependent attributes, time-series or sequence based modelling paradigms like LSTM can be used to improve the model to identify long term drifts in the users' behaviour patterns.

Last but not the least, testing the proposed framework in operational environments of various organizations with dissimilar settings and security concerns will be useful in determining real life feasibility of the proposed framework. Standardizing and enhancing the system's functionality regarding hypothesized and actual large-scale large and low-resource applications will be pertinent. Integration of the proposed system with other research studies and existing UEBA tools may create a foundation for more accurate, timely, and more automated insider threat detection solutions that could address the modern and rapidly increasing threats faced in organizations.

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